NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

NTU Academy for Professional and Continuing Education

(SCTP) Advanced Professional Certificate

Data Science and AI

NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

3.6 Time Series Data & Forecasting

Module Overview

3.1 Probability and Statistics for Machine Learning

3.2 Introduction to Machine Learning

3.3 Supervised Learning

3.4 Supervised Learning - Advanced

3.5 Unsupervised Learning

3.6 Time Series Data & Forecasting

3.7 Neural Network & Deep Learning

3.8 Computer Vision (CV)

3.9 Natural Language Processing (NLP)

3.10 NLP - Advanced

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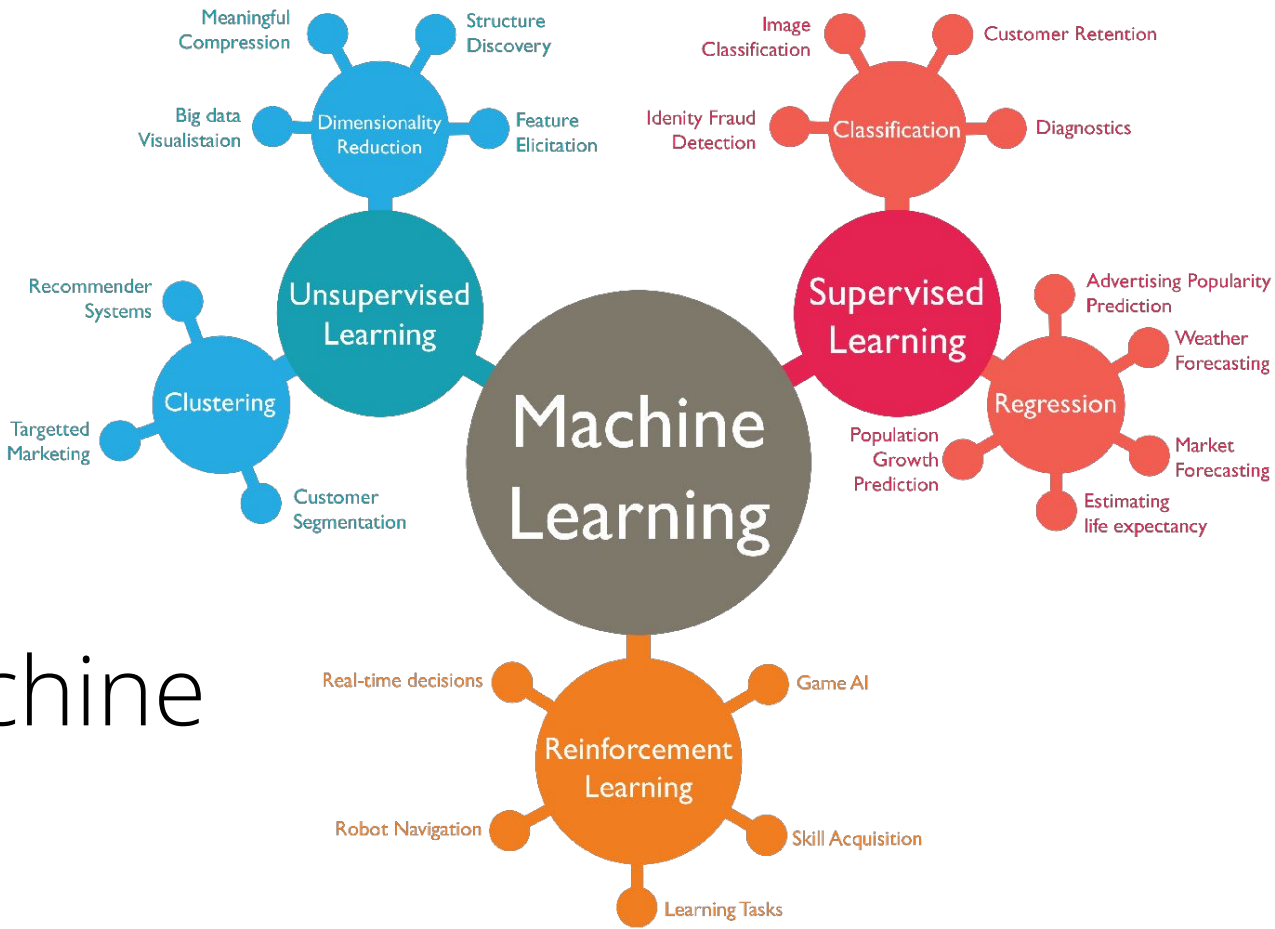
Types of Machine Learning



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Lesson Objectives

- Plot various time-series charts
- Perform time-series decomposition
- Perform time-series forecasting

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Lesson Plan

- **Time Series Data**
- Time Series Visualisation
- Time Series Analysis
- Forecasting Analysis

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What is Time Series?

Time series analysis is used when you have data that is collected over time and want to analyze patterns, trends, or make predictions about future values based on historical data. You typically use time series analysis when:

- **Forecasting:** Predicting future values based on past data, such as stock prices, weather patterns, or sales figures.
- **Trend analysis:** Identifying long-term movements or trends in data, such as overall growth or decline in economic indicators, consumer behavior, or company performance.
- **Seasonality:** Understanding and forecasting patterns that repeat at regular intervals, like monthly sales spikes during holidays, daily traffic patterns, or quarterly financial cycles.

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What is Time Series?

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- **Anomaly detection:** Detecting unusual patterns or outliers in data over time, which could indicate fraud, errors, or significant changes in behavior, like a sudden drop in website traffic.
- **Causal impact:** Investigating the effect of external events or interventions on a time series, such as the impact of a marketing campaign on sales or changes in government policy on unemployment.
- **Cycle analysis:** Identifying and modeling recurring cycles that aren't necessarily fixed, like economic booms and recessions, or longer-term cycles in industries.

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Time Series Data

- Time series data is data that is recorded over consistent intervals of time.
- Key characteristic: Sequential Nature - Each data point is dependent on the previous points, meaning time order matters.
- Examples:
  - **Financial**: Stock market prices, cryptocurrency trends
  - **Economics**: GDP growth, inflation rates
  - **Business**: Sales revenue, website traffic
  - **Science**: Temperature readings, climate patterns
  - **Healthcare**: Patient vital signs, disease progression

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Time Series Data

Pattern Type	Definition	Regularity	Cause	Predictability	Example
Trend	Long-term increase or decrease in the data	Generally gradual and consistent	Underlying structural changes (e.g., growth, decline)	Moderately predictable	Rising global temperatures over decades
Seasonal	Repeating short-term pattern at fixed, regular intervals	Regular (e.g., daily, monthly, yearly)	Calendar-based (e.g., weather, holidays)	Highly predictable	Retail spikes in December every year
Cyclic	Long-term fluctuations that occur at irregular intervals	Irregular	Economic or behavioral forces (not calendar-based)	Less predictable	Business cycles like booms and recessions

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Types of Time Series Data

- Time Series Data
- Cross Section Data
- Panel Data

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Time Series Data

Time series data consists of **sequential observations** recorded at **regular time intervals** for a single entity. The order of the observations is crucial since past values influence future values.

Examples:

- 📈 **Stock Market:** Daily closing prices of a stock.
- 🌡️ **Weather Data:** Hourly temperature readings.
- 🛒 **Sales Data:** Monthly sales of a specific product.

Date	Stock Price (USD)
10/3/2025	106.98
11/3/2025	108.86
12/3/2025	115.74
13/3/2025	115.58

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Cross Section Data

Cross-sectional data represents observations collected **at a single point in time** across multiple entities. It does not account for temporal changes. Time is NOT a factor.

Examples:

- **Stock Prices:** Daily closing prices of across different stocks.
- **Income Survey:** Household incomes of 1,000 people in 2024.
- **Demographics:** Educational levels of students across different cities.
- **COVID-19 Cases:** Number of infections per country as of June 2024.

Date	NVIDIA	TESLA	AMAZON
10/3/2025	106.98	222.15	194.54

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Panel Data

Panel data (or **longitudinal data**) tracks **multiple entities** over **multiple time periods**. It combines aspects of both **time series** and **cross-sectional** data.

Examples:

- **Stock Prices:** Daily closing prices of different stocks in the past 4 days.
- **Economic Growth:** GDP of multiple countries from 2000 to 2024.
- **Employee Salaries:** Annual salaries of employees from different companies from 2015 to 2024.

Date	NVIDIA	TESLA	AMAZON
10/3/2025	106.98	222.15	194.54
11/3/2025	108.86	230.48	196.59
12/3/2025	115.74	248.09	198.89
13/3/2025	115.58	240.68	193.89

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How to Choose the Right Data Type for Your Analysis?

- If you are studying **trends over time** → Use **Time Series Data**.
- If you are comparing **different entities at a single point in time** → Use **Cross-Sectional Data**.
- If you are analyzing **both temporal and cross-sectional patterns** → Use **Panel Data**.

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Lesson Plan

- Time Series Data
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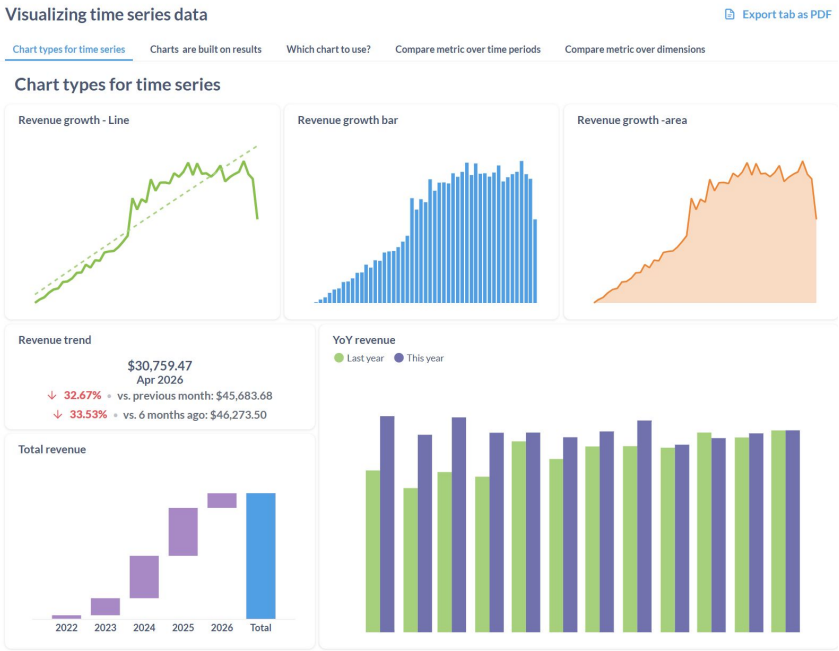
Time Series Data Visualisation

Visualizing time series data helps to:

- **Identify Trends**: Discover long-term upward or downward movements in data.
- **Understand Seasonality**: Detect periodic patterns like sales spikes during holidays.
- **Spot Anomalies**: Highlight unexpected changes, such as outliers or unusual trends.
- **Communicate Insights**: Share findings effectively with stakeholders.

[Example]



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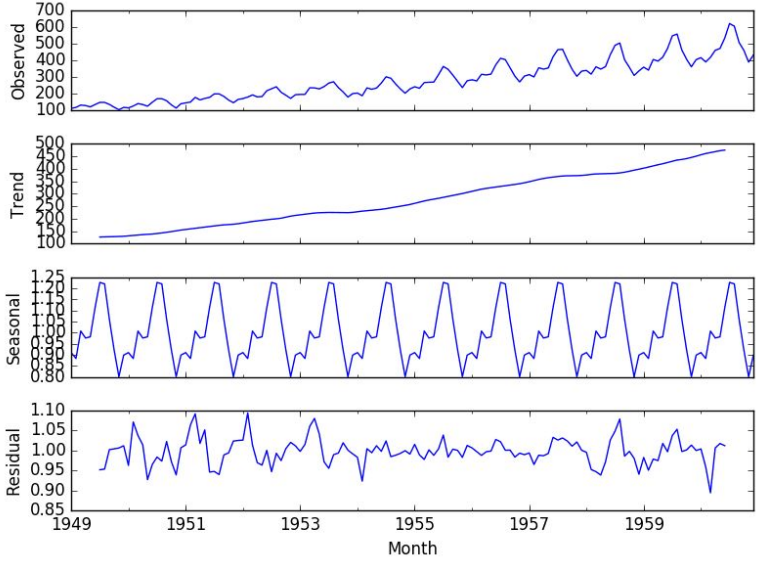
Time Series Decomposition

- Time series decomposition is essentially a process of deconstructing a time series into its constituent parts.
  - **Trend** (long-term direction; increasing / decreasing)
  - **Seasonality** (recurring patterns; monthly, season, etc)
  - **Irregular components** (random noise)
- Understand Patterns: Separates the effects of trend, seasonality, and noise.
- Modeling Readiness: Decomposed data is easier to model and analyze.



https://machinelearningmastery.com/decompose-time-series-data-trend-seasonality/

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Time Series Decomposition

2 key methods:

Additive

$$Y = T + S + R$$

- Consistent change on numerical scale
- Used when seasonal variation is independent of the trend.

Multiplicative

$$Y = T * S * R$$

- Change is on a relative scale
- Used when seasonal variation is proportional to the trend.

Key Considerations

- ✓ Use multiplicative when seasonal variation grows as the trend increases (peaks and troughs get bigger).
- ✓ Possible to change a multiplicative model to additive via log transformation / Box-Cox Transformation.

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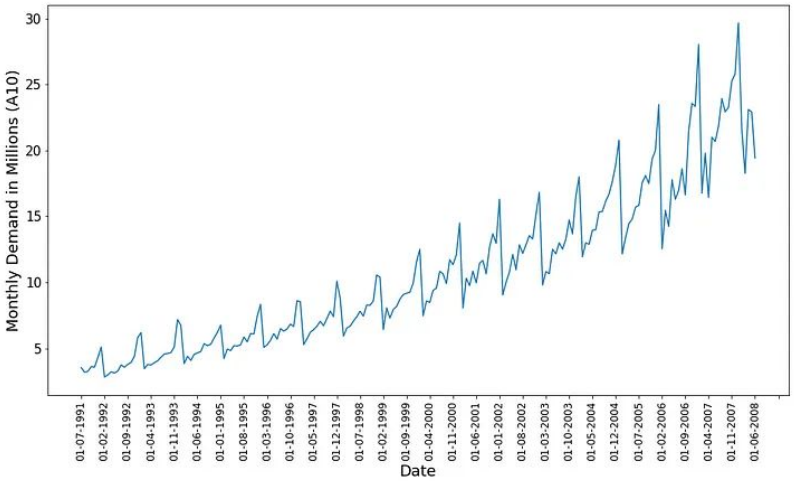
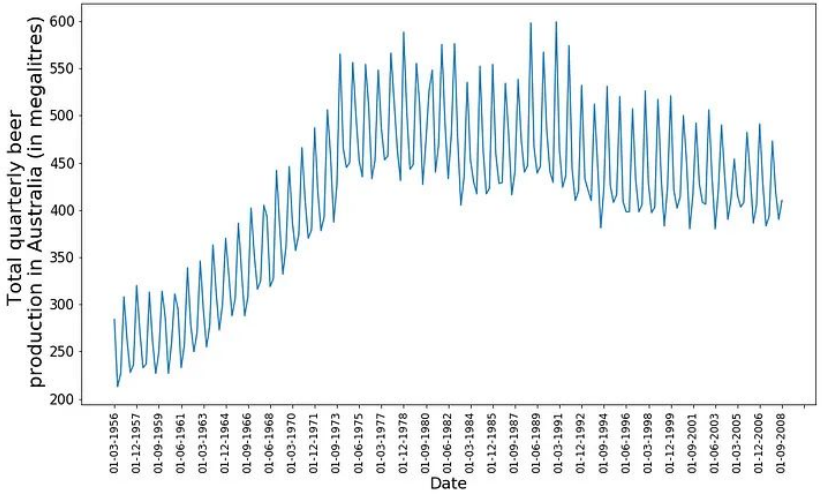
Additive? Multiplicative?





https://medium.com/analytics-vidhya/time-series-decomposition-part-i-trend-cycle-computation-29fac227896a#:~:text=Additive%20decomposition%20is%20generally%20used,is%20proportional%20to%20the%20trend.

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Time Series Decomposition

Approach (Classical Approach)

1. Compute Trend Component

using moving / rolling average

2. De-trend the Series

(Y - T) or (Y / T)

3. Compute Seasonal Component

by taking the average of de-trended series for each season

4. Calculate Residual Component

(Y - T – S) or (Y / (T * S))

Other approaches – STL, X11 , SEATS

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Code-along

Jupyter Notebook

Part 1: Visualizing Time-Series Data

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Code-along

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Part 2: Time Series Decomposition

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Part 3: Time-Series Forecasting Evaluation and Metrics

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Lesson Plan

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Time Series Forecasting Approaches

- AutoRegression (AR)
- Moving Average (MA)
- Autoregressive Integrated Moving Average (ARIMA)

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Why Stationarity?

Changing mean and variance over time $\rightarrow$ Changing distribution over time, which makes it harder to model

Stationarity means we can assume there is a universal distribution across the data which can be modelled, so it will be easier to forecast and predict.

1. No long term trend
2. No obvious seasonality
3. Constant Mean
4. Constant variance

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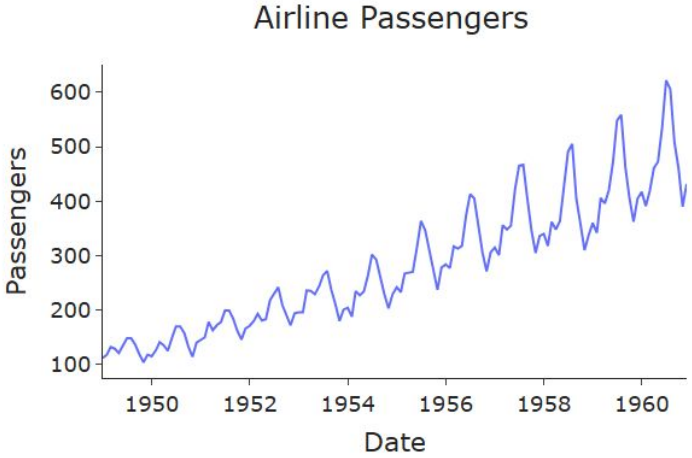
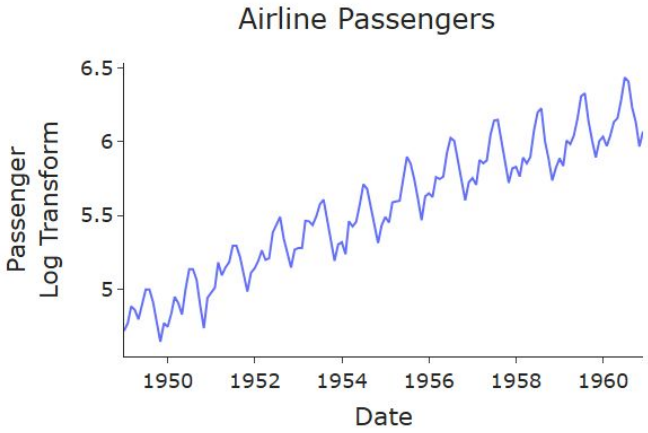
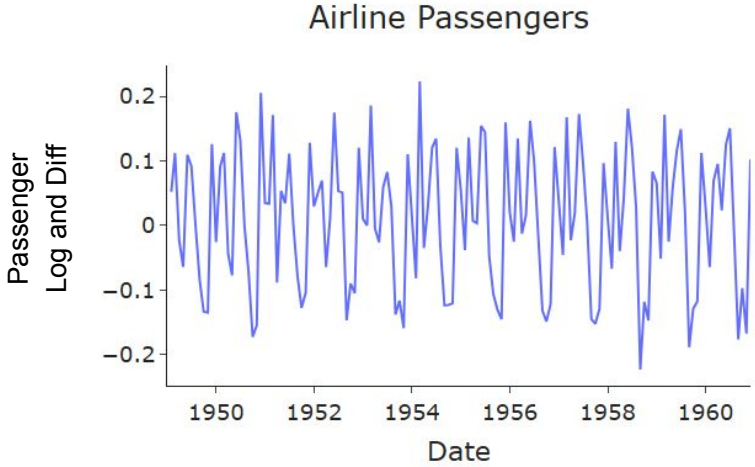
Stationarity







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Autoregressive Model (AR)

An Autoregressive model is a type of time series model where the current value of the series is expressed as a linear combination of its past values, plus some random noise.

$$y_t = c + \phi_1 y_{t-1} + \phi_2 y_{t-2} + \ldots + \phi_p y_{t-p} + \epsilon_t$$

Where:

- y is the time series forecasted at various time steps,
- ϕ are the fitted coefficients of the lags for the time series,
- ε is the error term (typically normally distributed)
- p is the number of lagged components included in the model, this is also known as the order.

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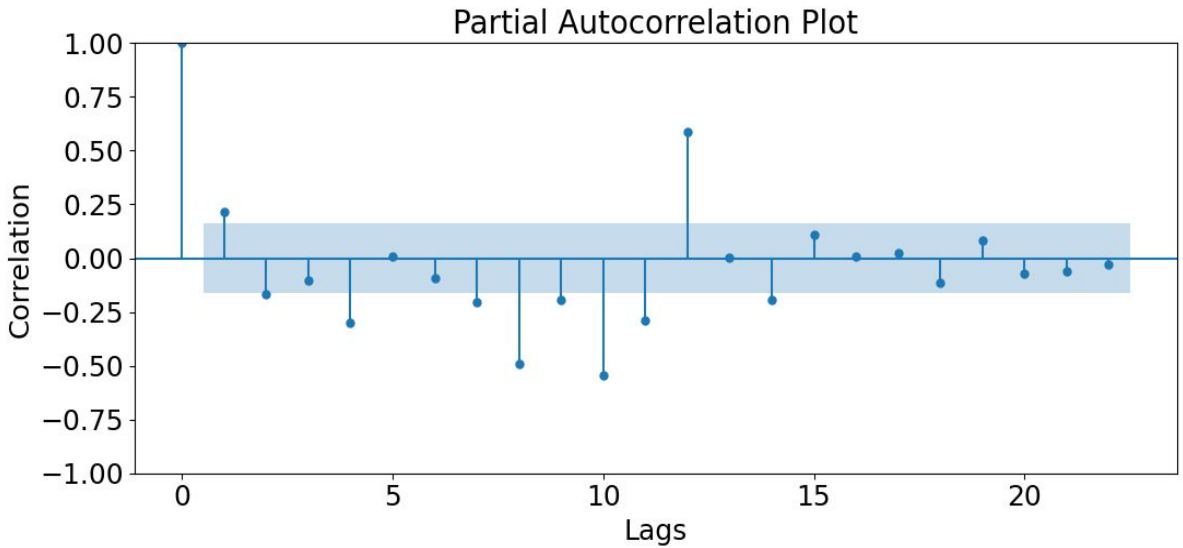
Autoregressive Order Selection - Partial AutoCorrelation



However, for building our model we will use the recommended approach of simply iterating over all the possible combinations of lags and choose the best model from that analysis.

- The lags outside the blue shaded region are classed as significant and should be included as features for our AR model.
- Lags 1, 2, 4, 7, 8, 9, 10, 11, 12 and 13 are significant.
- Lag 12 has the largest peak because the time series is indexed by month and has a yearly seasonality, hence lag 12 is an exact year difference.

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AR Interpreting the Coefficients

Positive ϕ	Series tends to follow the same direction (⬆️⬇️)
Negative ϕ	Series tends to reverse direction
$\lvert\phi\rvert < 1$	Series is stationary
$\lvert\phi\rvert \geq 1$	Series is non-stationary or explosive (bad!)

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Moving Average (MA)

- Not the same as a “simple moving average” that just averages past values.
- Instead, a Moving Average (MA) model is a time series model where the current value depends on past forecast errors (i.e., the white noise or residuals).

$$y_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1} + \theta_2 \epsilon_{t-2} + \ldots + \theta_q \epsilon_{t-q}$$

- Instead of using past values (like AR), the MA model uses past errors, which corrects for temporary noise or irregularities in the data.
- Not to be confused with the rolling mean model, which is also dubbed as a moving average model.

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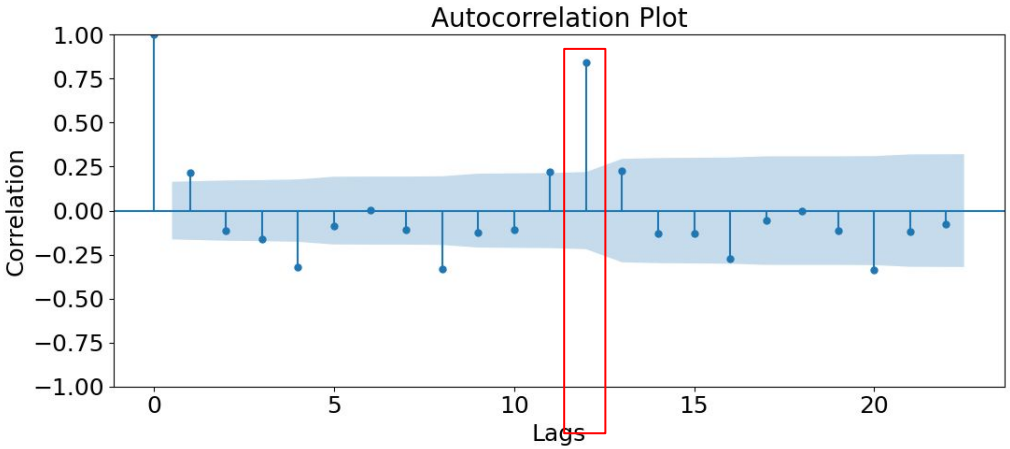
MA Order Selection - Autocorrelation

- Selecting the order of an MA model can be accomplished by viewing the autocorrelation function.
- Autocorrelation measures the correlation of the time series at various time steps (lags).
- If a lag has a high correlation, then it is influential in describing what the current value of the time series is.
- This idea of 'influence' can be used to carry out forecasts.



We can use autocorrelation to plot a *correlogram* of the various lags of the time series to determine which are statistically significant in impacting the forecasts.

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MA Interpreting the Coefficients

Let's say we have an MA(1) model:

$$X_t = \mu + \epsilon_t + \theta_1 \epsilon_{t-1}$$

The coefficient θ tells you how the previous time step error influences the current value

If θ_1 = 0.5

- A positive error yesterday (e.g., model underpredicted by +2) causes today's forecast to increase by 1 (0.5 × 2).
- A negative error yesterday (e.g., overpredicted by −2) would lower today's forecast by 1.

If θ_1 = -0.5

- A positive error yesterday causes a decrease today.
- A negative error yesterday causes an increase today.

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ARIMA Models

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Autoregressive Integrated Moving-Average (ARIMA)

ARIMA is a time series forecasting model that combines three components to model patterns in data over time — especially when there's trend or *autocorrelation* (i.e., when past values influence future values).

ARIMA is represented as ARIMA(p, d, q), where:

- p: No. of autoregressive (AR) terms.
- d: No. of differences needed to make the series stationary.
- q: No. of moving average (MA) terms.

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ARIMA Formula

The full ARIMA equation is just a linear summation of the three components.

Written in a shorthand way as ARIMA(p, d, q) where p, d and q refer to the order of autoregressive, differencing and moving-averages components respectively.

$$y'_t = c + \phi_1 y'_{t-1} + \ldots + \phi_p y'_{t-p} + \epsilon_t + \theta_1 \epsilon_1 + \ldots + \theta_q \epsilon_{t-q}$$

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ARIMA(p, d, q) Parameters

p: No. of autoregressive (AR) terms.

d: No. of differences needed to make the series stationary.

- Determined using repeated KPSS tests.
- Typically try d = 0, 1, 2. Avoid overdamping / noisy models

q: No. of moving average (MA) terms.

Once d is determined, p and q are then chosen by using ACF for MA(q) and PACF for AR(p)

AutoARIMA

Test different combinations of (p, d, q) and chooses the best one using AIC/BIC model selection criteria.

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ARIMA Coefficients Interpretation

Coeff. Type	What It Represents	Interpretation
AR (Autoregressive)	The relationship between past values and the current value	A positive coefficient means the series moves in the same direction as the past value; a negative coefficient means the opposite.
MA (Moving Average)	The influence of past forecast errors (residuals) on the current value	A positive coefficient means past errors push the series in the same direction, while a negative coefficient means they push in the opposite direction.
Constant (Intercept)	The baseline level of the time series after accounting for trends and seasonality	Represents the average value when there's no trend or autocorrelation.
Differencing (d)	The number of times the data has been differenced to remove trends	No direct interpretation, but it's important for making the series stationary.

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ARIMA Advantages

Advantage	Description
📈 Handles Trends	ARIMA can model and forecast time series with trends by differencing the data (d).
🔄 Flexible	Works for a wide variety of time series data (univariate), including those with trend or autocorrelation.
📚 Well-established and Supported	ARIMA is widely used and supported, with extensive documentation and tools like statsmodels and pmdarima.
⏳ Good for Short-Term Forecasting	Ideal for short-to-medium term forecasting, especially when there’s trend or autocorrelation.
🧠 Model Interpretability	The model structure is easy to interpret and provides insights into how past values and errors influence future predictions.
🛠️ Widely Supported in Software	Tools like Python's statsmodels and R make it easy to implement and tune ARIMA models.

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ARIMA Disadvantages

Limitation	Description
🚫 Requires Stationarity	ARIMA requires the data to be stationary, meaning the mean, variance, and autocorrelation should not change over time. Preprocessing is needed to achieve this (e.g., differencing).
⚠️ Limited to Linear Relationships	Assumes future values are linear combinations of past values — it can't capture nonlinear patterns.
🏃 Short Memory	ARIMA models only look back a fixed number of lags (p) — it may not capture long-term dependencies unless the order of p is large.
❓ Model Order Selection Can Be Tricky	Choosing the right parameters (p, d, q) can be subjective and require model selection methods like AIC/BIC.
🌪️ Poor Performance with Structural Breaks or Shocks	Struggles when there are sudden changes, structural breaks, or outliers (e.g., financial crises or natural disasters).
📢 Sensitive to Noise and Outliers	ARIMA models can be heavily influenced by outliers, particularly in small datasets.

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Forecasting Case Study

- ARIMA was used to forecast the number of new COVID-19 cases, deaths, and recoveries based on the daily reported data from different countries for assessment of the future outbreak

  Source: https://bmcmedresmethodol.biomedcentral.com/articles/10.1186/s12874-022-01755-x

- Uber

  Source: https://www.uber.com/en-SG/blog/forecasting-introduction/

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Code-along

Jupyter Notebook

Part 4: Forecasting Models (ARIMA)

+ Class Activity

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